EDA_Brian

Brian Park

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Loading Libraries

```
library(tidyverse)
library(lubridate)
library(ggplot2)
library(dplyr)
library(corrplot)
library(here)
library(scales)
library(forcats)
```

Read from current data version

```
df <- read.csv(here("data", "processed", "reviews_with_sent_scores.csv"))

dim(df)
```

```
## [1] 1000   76
```

```
head(df)
```

```
##   review_id rating_date count_helpful count_unhelpful employee_job_end_year
## 1  28951745  2019-08-30            10               0                  2019
## 2  59978917  2022-02-19             2               0                    NA
## 3  42749601  2021-02-19             0               0                  2021
## 4  12095563  2016-09-27             4               0                  2015
## 5  46084540  2021-04-26             0               0                    NA
## 6  58316203  2022-01-21             1               0                  2021
##   employee_length employee_status covid_related_review flags_business_outlook
## 1              20         REGULAR                    0               NEGATIVE
## 2               6         REGULAR                    0               POSITIVE
## 3               2         REGULAR                    1               POSITIVE
## 4               2         REGULAR                    0               NEGATIVE
## 5               2         REGULAR                    1               NEGATIVE
## 6               2                                    0               POSITIVE
##   flags_ceo_approval flags_recommend_frend rating_culture_values
```

```
## 1         DISAPPROVE              NEGATIVE                     1
## 2            APPROVE              POSITIVE                     3
## 3            APPROVE              POSITIVE                     5
## 4         NO_OPINION              NEGATIVE                     1
## 5            APPROVE              NEGATIVE                     5
## 6            APPROVE              POSITIVE                     5
##   rating_diversity_inclusion rating_overall rating_work_life
## 1                          3              2                1
## 2                          5              5                2
## 3                          5              4                5
## 4                          5              2                1
## 5                          5              3                3
## 6                          5              5                5
##                                                                      summary
## 1                                                        IT is Out of Control
## 2                                                  Great Company to work with
## 3                                                                       Great
## 4 Vodafone is an Egotistical Company that only cares about its net promoter score
## 5                                 Great company, poor processes, high expectations
## 6                                                  Commercial Product Internship
##      company_name   employee_location       employee_job_title
## 1 Elevance Health                                    Manager
## 2       HDFC Bank           New Delhi Assistant Vice President
## 3          Google             Chennai       Software Developer
## 4        Vodafone     London, England     Retail Store Manager
## 5         Verizon Rolling Meadows, IL Business Account Manager
## 6    Florida Blue
##
## 1                              Humble yourselves. Remember when you were coming up in your career. R
## 2
## 3
## 4  You have some amazing staff members who are totally disengaged. Your TNPS system is flawed. Your
## 5 Handle customer escalations so BAMs can focus on sales. Be realistic about sales goals, especially
## 6
##
## 1                                            Salary was good for the position, but only because the
## 2                Great brand name. Top Management are consistent on bank's goals and strategies. Con
## 3                                                                                          Good a
## 4                                                   Unfortunately there were no pros during my time w
## 5 Working with business owners or employees to solve telecommunication and technology. Great salary,
## 6                                  Very inclusive and helpful environment which made succeeding in t
##
## 1
## 2
## 3
## 4                                        Retail stores are severely understaffed. The new system, S
## 5 Sales expectations are unrealistic. Due to Covid, the job shifted from outside sales to inside sal
## 6
##   rating_compensation_benefits rating_senior_leadership
## 1                            2                       1
## 2                            3                       5
## 3                            5                       5
## 4                            2                       2
## 5                            4                       2
```

```
## 6                           5                        5
##   rating_career_opportunities             sector rating_year helpfulness_score
## 1                           1         Healthcare        2019                10
## 2                           5            Finance        2022                 2
## 3                           3         Technology        2021                 0
## 4                           2 Telecommunications        2016                 4
## 5                           3 Telecommunications        2021                 0
## 6                           3            Finance        2022                 1
##   career_missing employee_job_quit employee_job_start_year has_review_text
## 1          FALSE              TRUE                    1999            TRUE
## 2          FALSE             FALSE                    2016            TRUE
## 3          FALSE              TRUE                    2019            TRUE
## 4          FALSE              TRUE                    2013            TRUE
## 5          FALSE             FALSE                    2019            TRUE
## 6           TRUE              TRUE                    2019            TRUE
##   distilbert_summary_label distilbert_summary_score distilbert_pros_label
## 1                 NEGATIVE                0.9988170              NEGATIVE
## 2                 POSITIVE                0.9998851              POSITIVE
## 3                 POSITIVE                0.9998630              POSITIVE
## 4                 NEGATIVE                0.9995903              NEGATIVE
## 5                 NEGATIVE                0.7545955              POSITIVE
## 6                 NEGATIVE                0.9400896              POSITIVE
##   distilbert_pros_score distilbert_cons_label distilbert_cons_score
## 1             0.9892550              NEGATIVE             0.9951473
## 2             0.9998654              NEGATIVE             0.9947620
## 3             0.9998626              POSITIVE             0.9993190
## 4             0.9979458              NEGATIVE             0.9991649
## 5             0.9741599              NEGATIVE             0.9978154
## 6             0.9997764              POSITIVE             0.7481209
##   distilbert_advice_label distilbert_advice_score distilbert_summary_signed
## 1                POSITIVE               0.9990497                -0.9988170
## 2                POSITIVE               0.9998141                 0.9998851
## 3                POSITIVE               0.9998161                 0.9998630
## 4                NEGATIVE               0.9995571                -0.9995903
## 5                NEGATIVE               0.7379106                -0.7545955
## 6                POSITIVE               0.7481209                -0.9400896
##   distilbert_pros_signed distilbert_cons_signed distilbert_advice_signed
## 1             -0.9892550             -0.9951473                0.9990497
## 2              0.9998654             -0.9947620                0.9998141
## 3              0.9998626              0.9993190                0.9998161
## 4             -0.9979458             -0.9991649               -0.9995571
## 5              0.9741599             -0.9978154               -0.7379106
## 6              0.9997764              0.7481209                0.7481209
##   roberta_summary_label roberta_summary_score roberta_pros_label
## 1              negative             0.8464990            neutral
## 2              positive             0.9722827           positive
## 3              positive             0.7565586           positive
## 4              negative             0.8649106           negative
## 5              positive             0.6938233           positive
## 6               neutral             0.7955984           positive
##   roberta_pros_score roberta_cons_label roberta_cons_score roberta_advice_label
## 1          0.5495484           negative          0.7179754              neutral
## 2          0.9709563           negative          0.6573127             positive
## 3          0.9774164           negative          0.5272856             positive
```

```
## 4          0.8827037           negative          0.8609493             negative
## 5          0.8999880           negative          0.8552947              neutral
## 6          0.9773394           positive          0.4012986             positive
##   roberta_advice_score roberta_summary_signed roberta_pros_signed
## 1            0.5416139             -0.8464990           0.0000000
## 2            0.9413118              0.9722827           0.9709563
## 3            0.6844373              0.7565586           0.9774164
## 4            0.9532588             -0.8649106          -0.8827037
## 5            0.6276173              0.6938233           0.8999880
## 6            0.4012986              0.0000000           0.9773394
##   roberta_cons_signed roberta_advice_signed vader_summary_neg vader_summary_neu
## 1          -0.7179754             0.0000000             0.000             1.000
## 2          -0.6573127             0.9413118             0.000             0.494
## 3          -0.5272856             0.6844373             0.000             0.000
## 4          -0.8609493            -0.9532588             0.119             0.692
## 5          -0.8552947             0.0000000             0.277             0.357
## 6           0.4012986             0.4012986             0.000             1.000
##   vader_summary_pos vader_summary_compound vader_pros_neg vader_pros_neu
## 1             0.000                 0.0000          0.188          0.690
## 2             0.506                 0.6249          0.000          0.612
## 3             1.000                 0.6249          0.000          0.210
## 4             0.189                 0.2732          0.315          0.685
## 5             0.366                 0.2500          0.000          0.731
## 6             0.000                 0.0000          0.000          0.530
##   vader_pros_pos vader_pros_compound vader_cons_neg vader_cons_neu
## 1          0.122             -0.2612          0.076          0.625
## 2          0.388              0.7650          0.090          0.805
## 3          0.790              0.9062          0.000          0.369
## 4          0.000             -0.5574          0.059          0.809
## 5          0.269              0.7096          0.152          0.808
## 6          0.470              0.8341          0.000          0.000
##   vader_cons_pos vader_cons_compound vader_advice_neg vader_advice_neu
## 1          0.299              0.6597            0.053            0.747
## 2          0.105             -0.1280            0.000            0.719
## 3          0.631              0.8201            0.000            0.000
## 4          0.132              0.7650            0.109            0.812
## 5          0.039             -0.7597            0.139            0.861
## 6          0.000              0.0000            0.000            0.000
##   vader_advice_pos vader_advice_compound summary_wc pros_wc cons_wc advice_wc
## 1            0.200                0.7269          5      13      23        35
## 2            0.281                0.8481          5      15      29        23
## 3            1.000                0.4404          1       6       8         1
## 4            0.079               -0.1053         13      12      79        42
## 5            0.000               -0.7003          6      18      75        38
## 6            0.000                0.0000          3      13       0         0
```

To make plots more appealing

```
theme_set(
  theme_minimal(base_size = 13) +
    theme(
```

```r
      plot.title = element_text(face = "bold", size = 15),
      plot.subtitle = element_text(size = 11),
      axis.title = element_text(face = "bold"),
      panel.grid.minor = element_blank(),
      legend.position = "bottom"
    )
)
```

1. Missing Data Analysis

```r
colSums(is.na(df))
```

```
##                    review_id                  rating_date
##                            0                            0
##               count_helpful              count_unhelpful
##                            0                            0
##        employee_job_end_year              employee_length
##                           49                            0
##              employee_status         covid_related_review
##                            0                            0
##       flags_business_outlook           flags_ceo_approval
##                            0                            0
##        flags_recommend_frend        rating_culture_values
##                            0                            0
##   rating_diversity_inclusion               rating_overall
##                            0                            0
##             rating_work_life                      summary
##                            0                            0
##                 company_name            employee_location
##                            0                            0
##           employee_job_title         advice_to_management
##                            0                            0
##                  review_pros                  review_cons
##                            0                            0
## rating_compensation_benefits     rating_senior_leadership
##                            0                            0
##  rating_career_opportunities                       sector
##                            0                            0
##                  rating_year            helpfulness_score
##                            0                            0
##               career_missing            employee_job_quit
##                            0                            0
##      employee_job_start_year              has_review_text
##                            0                            0
##     distilbert_summary_label     distilbert_summary_score
##                            0                            0
##        distilbert_pros_label        distilbert_pros_score
##                            0                            0
##        distilbert_cons_label        distilbert_cons_score
##                            0                            0
```

```
##      distilbert_advice_label      distilbert_advice_score
##                            0                            0
##    distilbert_summary_signed       distilbert_pros_signed
##                            0                            0
##       distilbert_cons_signed     distilbert_advice_signed
##                            0                            0
##        roberta_summary_label        roberta_summary_score
##                            0                            0
##           roberta_pros_label           roberta_pros_score
##                            0                            0
##           roberta_cons_label           roberta_cons_score
##                            0                            0
##         roberta_advice_label         roberta_advice_score
##                            0                            0
##       roberta_summary_signed          roberta_pros_signed
##                            0                            0
##          roberta_cons_signed        roberta_advice_signed
##                            0                            0
##            vader_summary_neg            vader_summary_neu
##                            0                            0
##            vader_summary_pos       vader_summary_compound
##                            0                            0
##               vader_pros_neg               vader_pros_neu
##                            0                            0
##               vader_pros_pos          vader_pros_compound
##                            0                            0
##               vader_cons_neg               vader_cons_neu
##                            0                            0
##               vader_cons_pos          vader_cons_compound
##                            0                            0
##             vader_advice_neg             vader_advice_neu
##                            0                            0
##             vader_advice_pos        vader_advice_compound
##                            0                            0
##                   summary_wc                      pros_wc
##                            0                            0
##                      cons_wc                    advice_wc
##                            0                            0
```

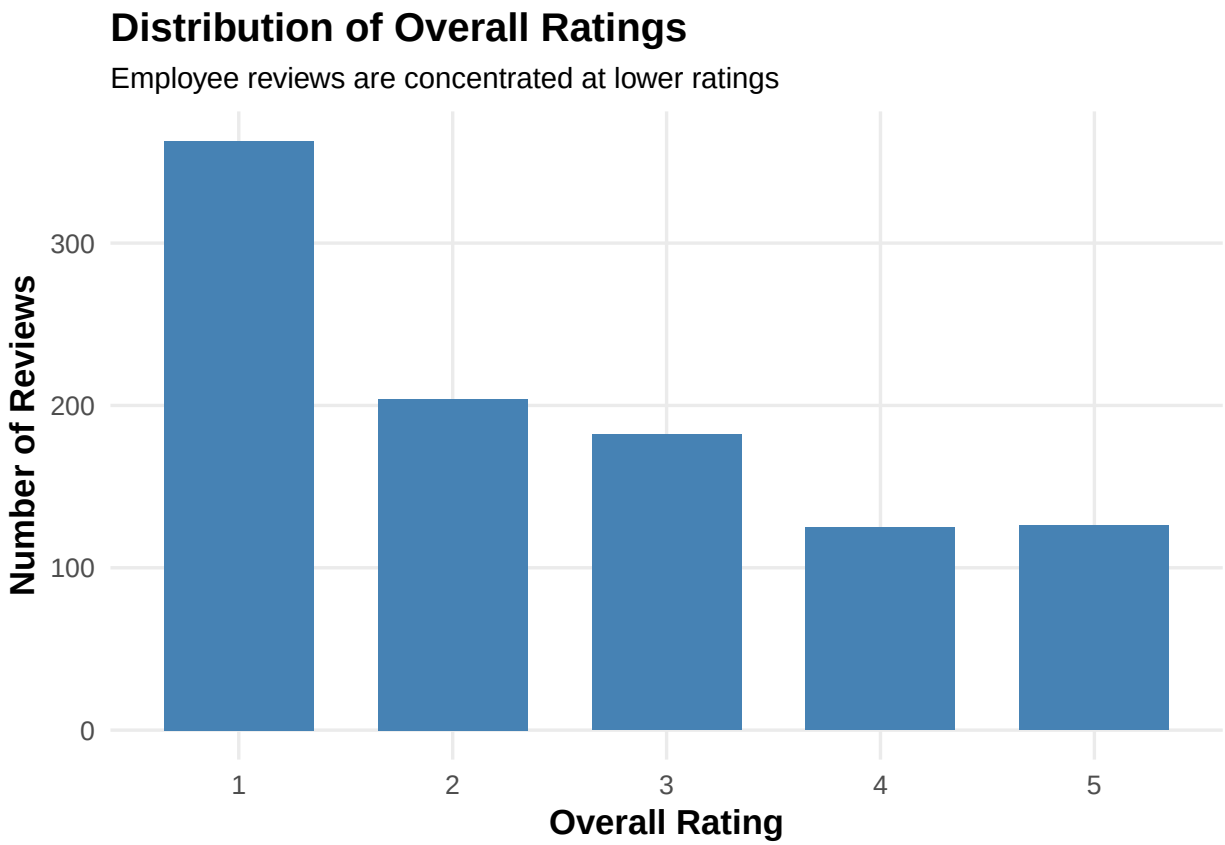
Explanation:

The variable employee_job_end_year contains missing values that correspond to current employees who have not left their jobs. These are structurally missing values and were intentionally preserved.

2. Overall Rating Distribution

```
ggplot(df, aes(x = factor(rating_overall))) +
  geom_bar(fill = "steelblue", width = 0.7) +
  labs(
    title = "Distribution of Overall Ratings",
    subtitle = "Employee reviews are concentrated at lower ratings",
```

```
    x = "Overall Rating",
    y = "Number of Reviews"
  )
```

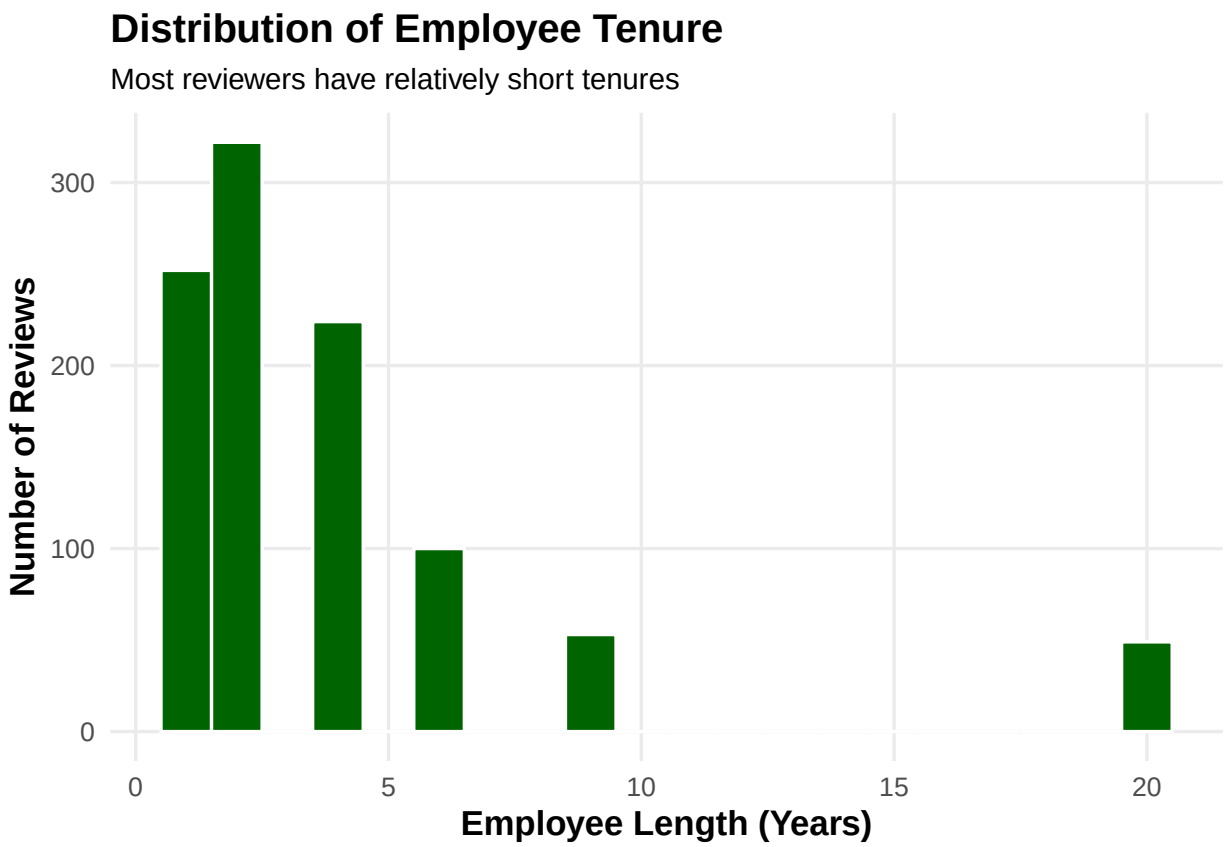


Explanation:

The distribution of overall ratings is skewed toward lower values, with a large number of reviews concentrated around ratings of 1 and 2. This indicates that negative reviews are more common in this dataset. Higher ratings (4–5) appear less frequently, suggesting that employees are more likely to report negative experiences than positive ones.

3. Employee Length Distribution

```
ggplot(df, aes(x = employee_length)) +
  geom_histogram(binwidth = 1, fill = "darkgreen", color = "white") +
  scale_x_continuous(breaks = pretty_breaks()) +
  labs(
    title = "Distribution of Employee Tenure",
    subtitle = "Most reviewers have relatively short tenures",
    x = "Employee Length (Years)",
    y = "Number of Reviews"
  )
```

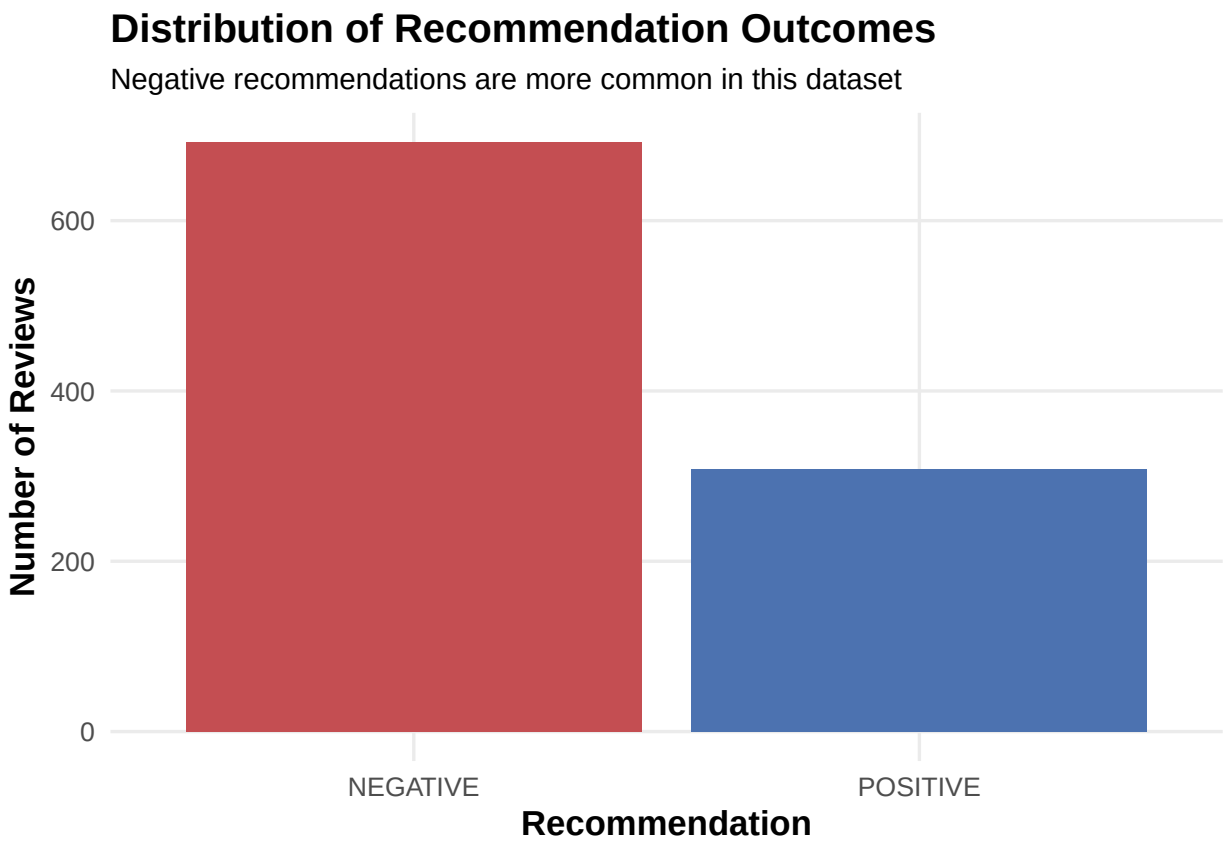


Explanation:

The distribution of employee tenure is right-skewed, with most employees having relatively short lengths of employment, typically between 1 and 3 years. As tenure increases, the number of employees decreases significantly. A small number of employees have long tenures, which appear as outliers in the distribution. This pattern suggests that employee turnover may be relatively high, with fewer individuals remaining at a company for extended periods.

4. Recommendation Distribution

```
ggplot(df, aes(x = flags_recommend_frend, fill = flags_recommend_frend)) +
  geom_bar(show.legend = FALSE) +
  scale_fill_manual(values = c("NEGATIVE" = "#C44E52", "POSITIVE" = "#4C72B0")) +
  labs(
    title = "Distribution of Recommendation Outcomes",
    subtitle = "Negative recommendations are more common in this dataset",
    x = "Recommendation",
    y = "Number of Reviews"
  )
```

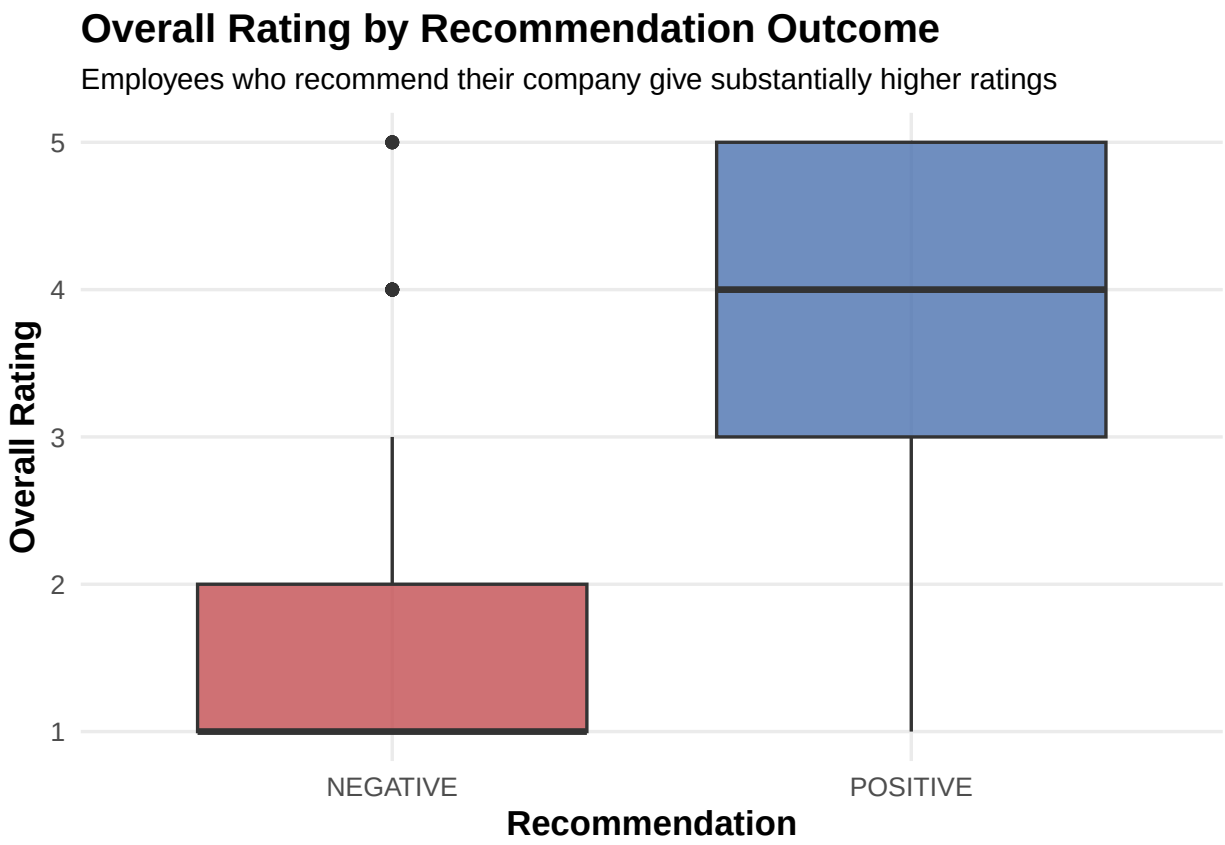


Explanation

The distribution of recommendation outcomes shows a clear imbalance, with significantly more negative recommendations than positive ones. This indicates that a larger proportion of employees in the dataset do not recommend their company. The imbalance suggests that negative experiences may be more frequently reported, which is consistent with the tendency for dissatisfied individuals to leave reviews more often than satisfied ones.

5. Overall Rating vs Recommendation

```
ggplot(df, aes(x = flags_recommend_frend, y = rating_overall, fill = flags_recommend_frend)) +
  geom_boxplot(alpha = 0.8, show.legend = FALSE) +
  scale_fill_manual(values = c("NEGATIVE" = "#C44E52", "POSITIVE" = "#4C72B0")) +
  labs(
    title = "Overall Rating by Recommendation Outcome",
    subtitle = "Employees who recommend their company give substantially higher ratings",
    x = "Recommendation",
    y = "Overall Rating"
  )
```



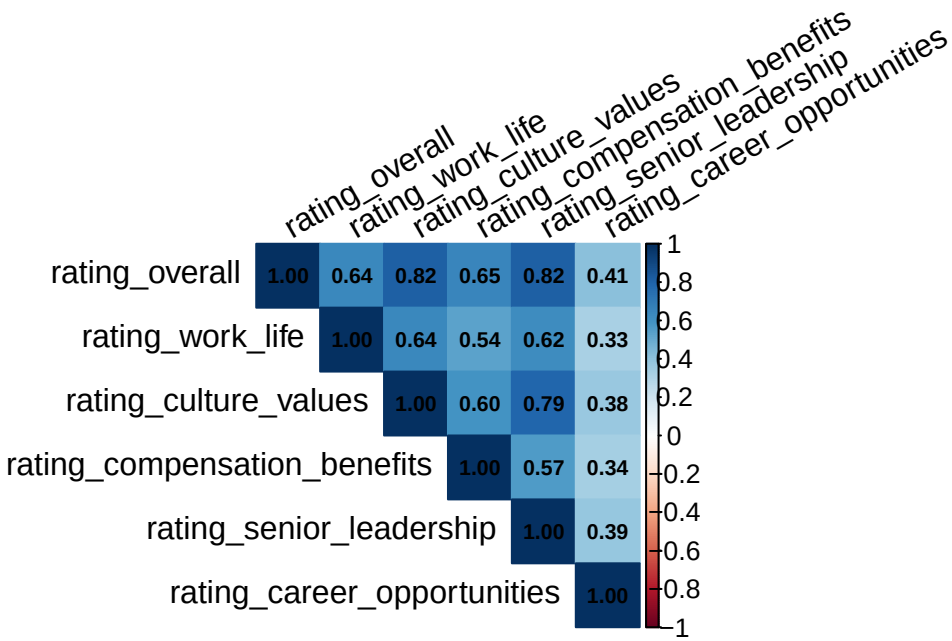
Explanation

There is a strong relationship between overall rating and recommendation behavior. Reviews with positive recommendations tend to have much higher overall ratings, while negative recommendations are associated with lower ratings. This suggests that overall rating is one of the strongest indicators of whether an employee would recommend the company.

6. Correlation Between Rating Variables

```
ratings <- df %>%
  select(
    rating_overall,
    rating_work_life,
    rating_culture_values,
    rating_compensation_benefits,
    rating_senior_leadership,
    rating_career_opportunities
  )

corr_matrix <- cor(ratings, use = "complete.obs")

corrplot(
```

```
  corr_matrix,
  method = "color",
  type = "upper",
  addCoef.col = "black",
  tl.col = "black",
  tl.srt = 30,
  number.cex = 0.7
)
```



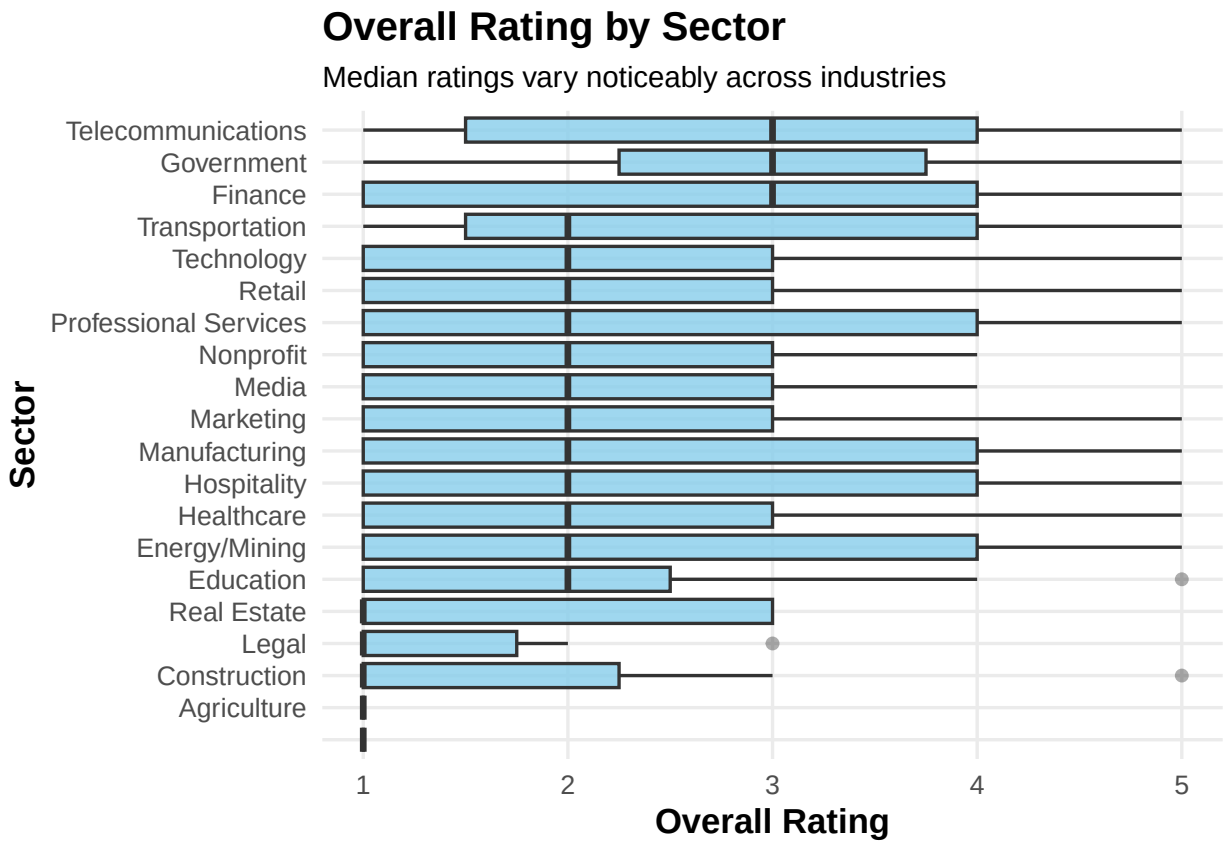
Explanation

The correlation matrix shows that the workplace rating variables are positively related to one another. In particular, overall rating tends to move together with culture, leadership, and career opportunities. This means that when employees rate one important workplace dimension highly, they often rate other dimensions highly as well, suggesting overlap among these measures.

7. Ratings by Sector

```
ggplot(df, aes(x = fct_reorder(sector, rating_overall, .fun = median), y = rating_overall)) +
  geom_boxplot(fill = "skyblue", alpha = 0.8, outlier.alpha = 0.4) +
  coord_flip() +
  labs(
```

```
    title = "Overall Rating by Sector",
    subtitle = "Median ratings vary noticeably across industries",
    x = "Sector",
    y = "Overall Rating"
  )
```



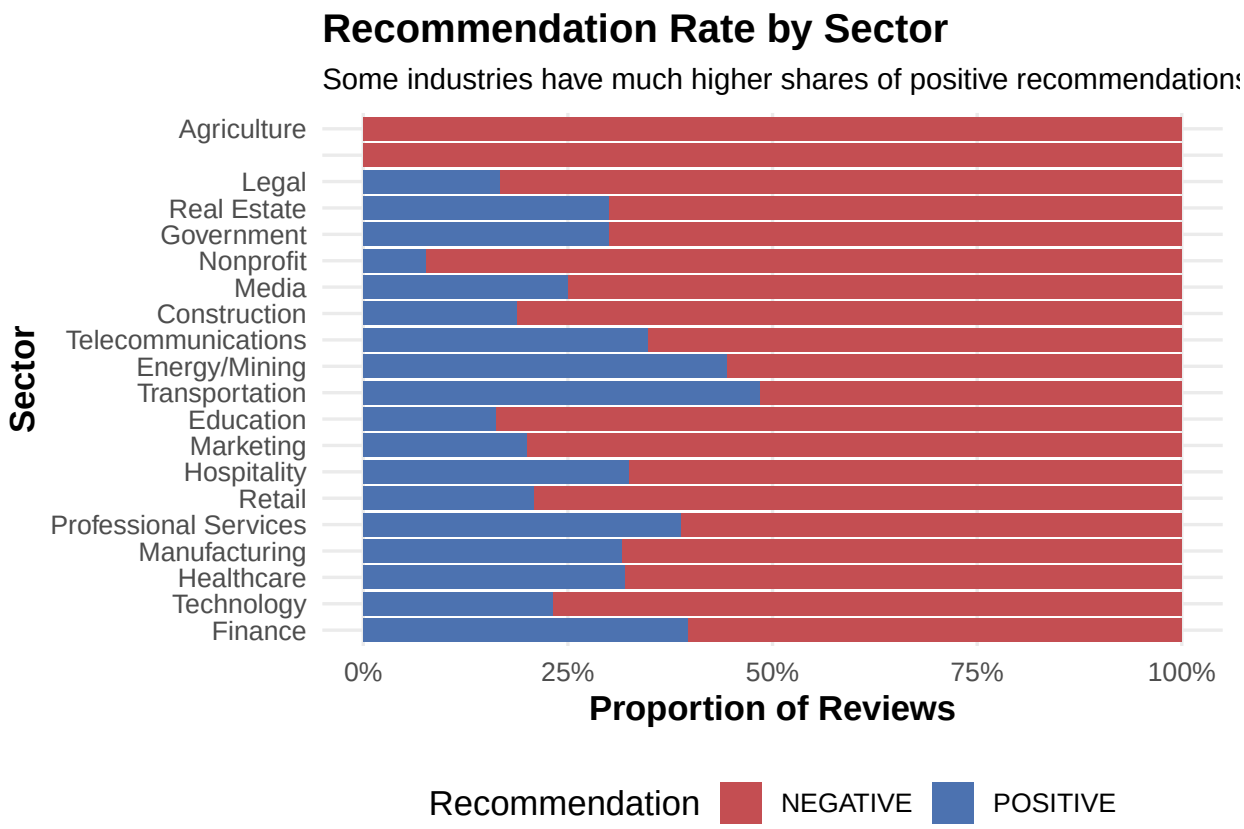
Explanation

Employee ratings vary across sectors, with some industries showing higher median satisfaction than others. The spread within each sector also differs, meaning that some sectors have more consistent employee experiences while others are more mixed. This suggests that industry context plays an important role in shaping employee satisfaction.

8. Recommendation Rate by Sector

```
ggplot(df, aes(x = fct_infreq(sector), fill = flags_recommend_frend)) +
  geom_bar(position = "fill") +
  coord_flip() +
  scale_y_continuous(labels = percent_format()) +
  scale_fill_manual(values = c("NEGATIVE" = "#C44E52", "POSITIVE" = "#4C72B0")) +
  labs(
    title = "Recommendation Rate by Sector",
```

```
    subtitle = "Some industries have much higher shares of positive recommendations",
    x = "Sector",
    y = "Proportion of Reviews",
    fill = "Recommendation"
  )
```



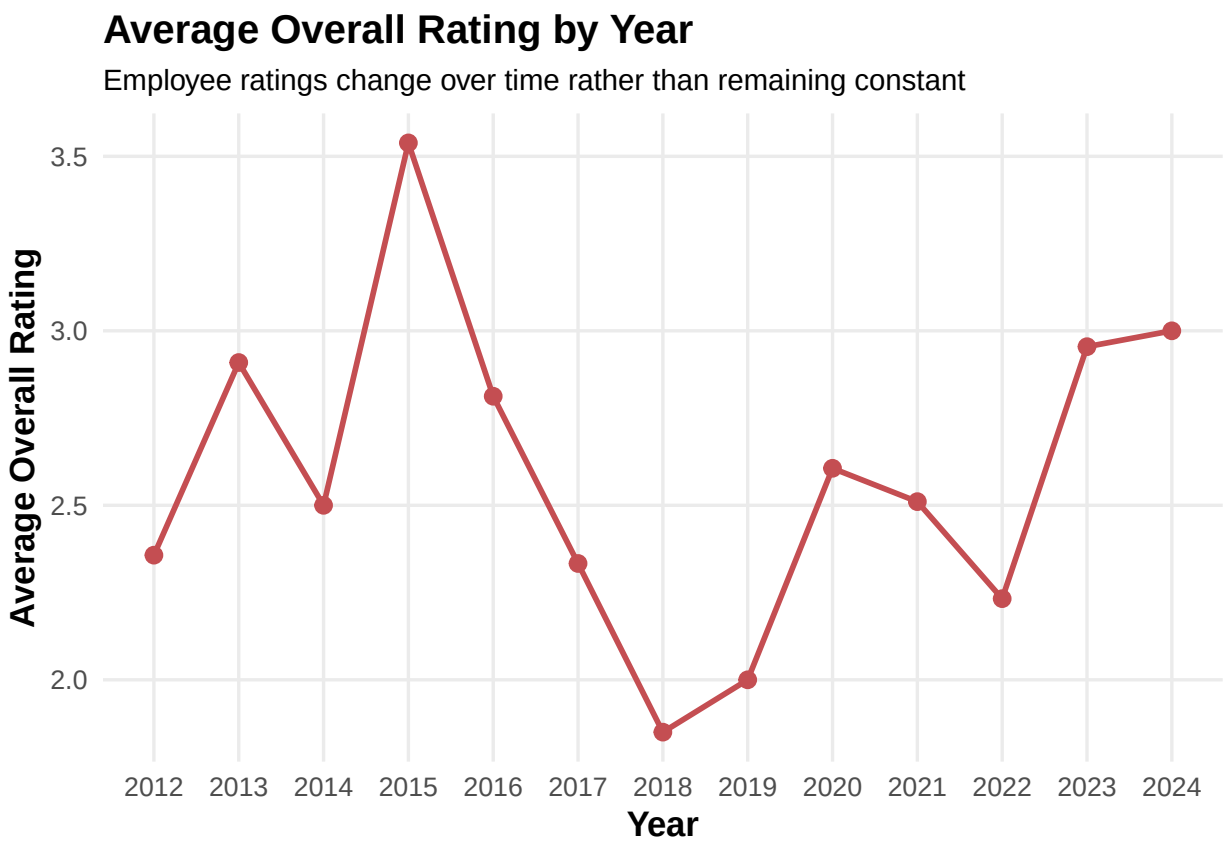
Explanation

The share of positive and negative recommendations differs across sectors. Some sectors have relatively balanced outcomes, while others are dominated by negative recommendations. This reinforces the earlier finding that industry matters and suggests that workplace satisfaction is not evenly distributed across the economy.

9. Average Overall Rating By Year

```
df$rating_date <- as.Date(df$rating_date)
df$rating_year <- as.numeric(format(df$rating_date, "%Y"))

yearly_avg <- df %>%
  group_by(rating_year) %>%
  summarise(avg_rating = mean(rating_overall, na.rm = TRUE), .groups = "drop")
```

```r
ggplot(yearly_avg, aes(x = rating_year, y = avg_rating)) +
  geom_line(color = "#C44E52", linewidth = 1) +
  geom_point(color = "#C44E52", size = 2.5) +
  scale_x_continuous(breaks = yearly_avg$rating_year) +
  labs(
    title = "Average Overall Rating by Year",
    subtitle = "Employee ratings change over time rather than remaining constant",
    x = "Year",
    y = "Average Overall Rating"
  )
```



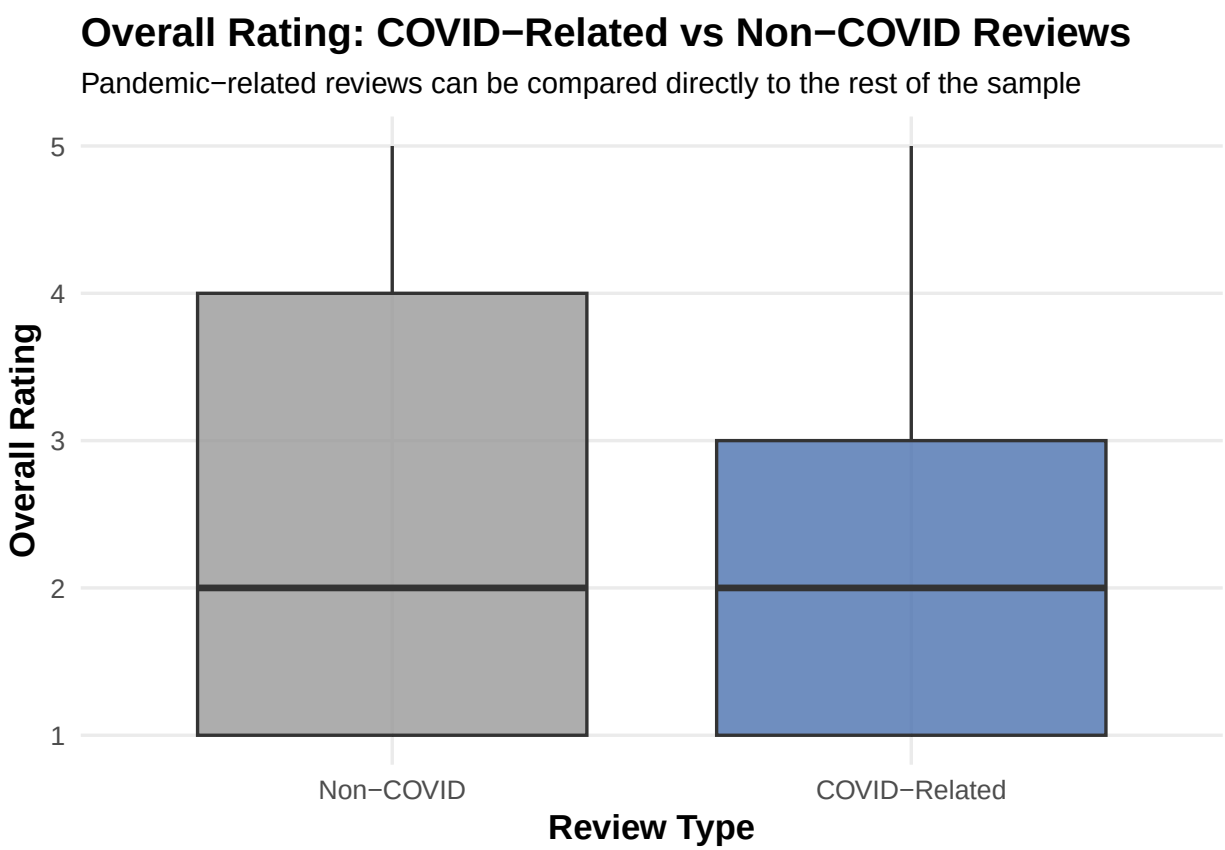
Explanation

Average ratings show a noticeable time trend rather than staying constant across years. The pattern suggests that employee sentiment changes over time, possibly reflecting shifts in workplace policies, labor-market conditions, or broader economic and social events.

10. COVID Impact Analysis

```r
ggplot(df, aes(x = factor(covid_related_review), y = rating_overall, fill = factor(covid_related_review
  geom_boxplot(show.legend = FALSE, alpha = 0.8) +
  scale_fill_manual(values = c("0" = "#999999", "1" = "#4C72B0")) +
```

```
  scale_x_discrete(labels = c("0" = "Non-COVID", "1" = "COVID-Related")) +
  labs(
    title = "Overall Rating: COVID-Related vs Non-COVID Reviews",
    subtitle = "Pandemic-related reviews can be compared directly to the rest of the sample",
    x = "Review Type",
    y = "Overall Rating"
  )
```



Explanation

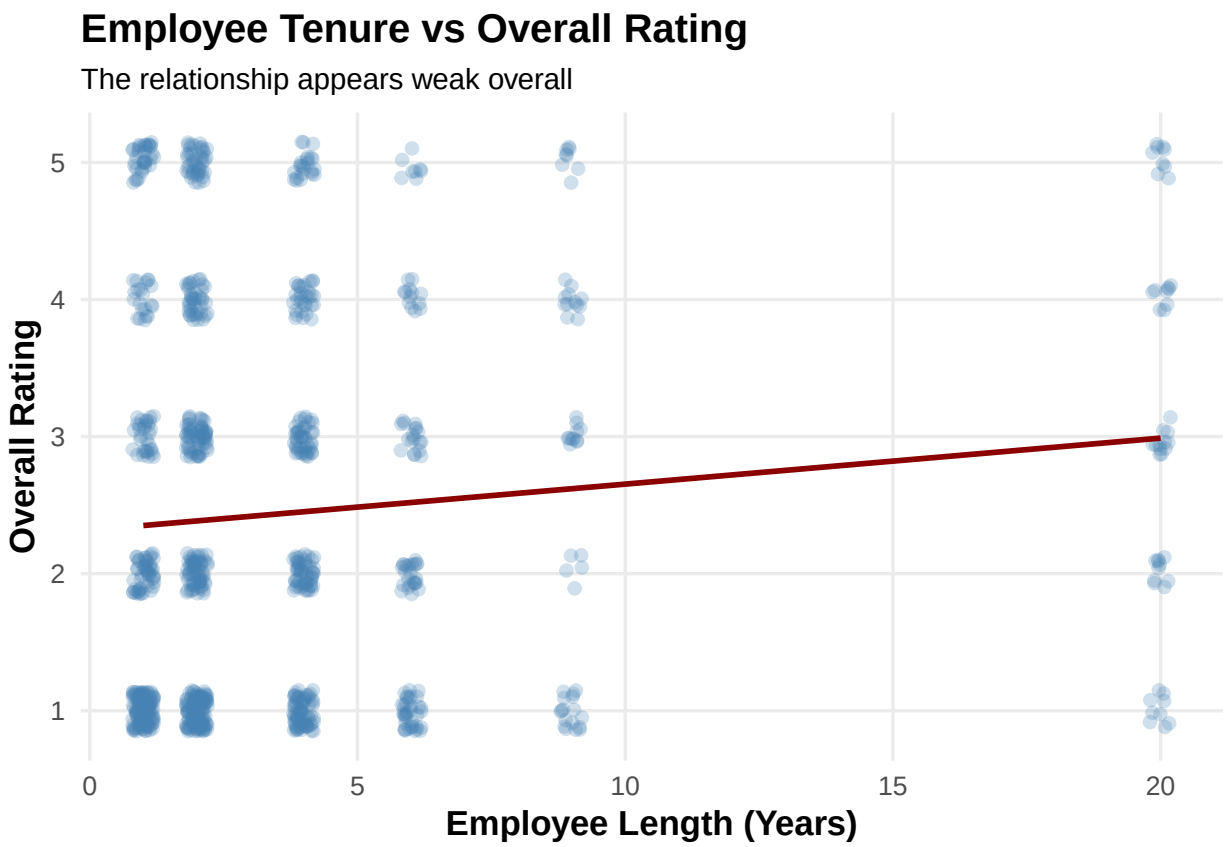
Comparing COVID-related reviews with non-COVID reviews helps show whether the pandemic period was associated with different employee experiences. Both have a median rating of approximately 2, indicating generally low employee satisfaction across the dataset. This suggests that the pandemic did not significantly change overall employee sentiment in this dataset, and that negative experiences were already common before COVID.

11. Employee Length vs. Rating

```
ggplot(df, aes(x = employee_length, y = rating_overall)) +
  geom_jitter(width = 0.2, height = 0.15, alpha = 0.25, color = "steelblue") +
  geom_smooth(method = "lm", se = FALSE, color = "darkred", linewidth = 1) +
  labs(
```

```
    title = "Employee Tenure vs Overall Rating",
    subtitle = "The relationship appears weak overall",
    x = "Employee Length (Years)",
    y = "Overall Rating"
  )
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



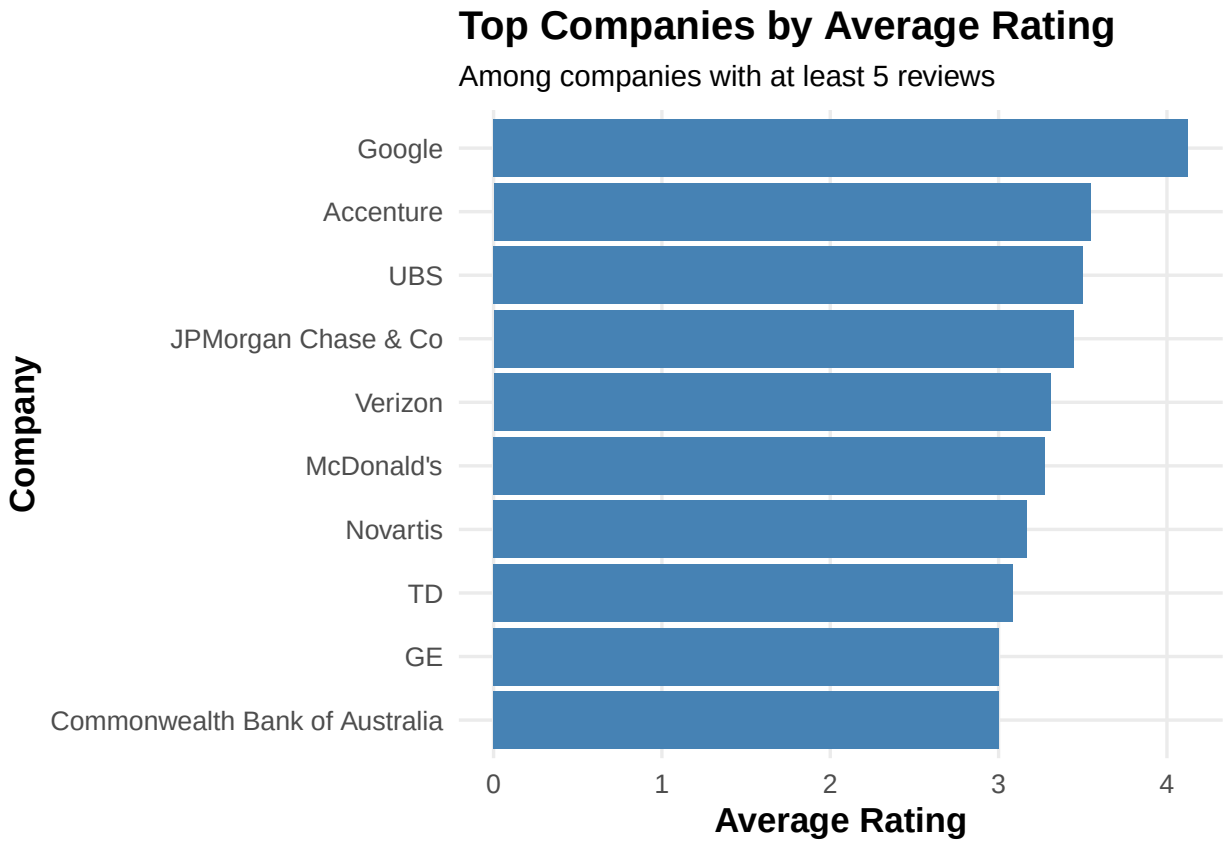
Explanation

The relationship between tenure and overall rating appears weak. Although there may be slight changes across different lengths of employment, employee satisfaction does not seem to be strongly determined by how long someone stayed at the company. This suggests that other factors, such as culture, management, or compensation, may matter more than tenure alone.

12. Top Companies By Rating

```
top_companies <- df %>%
  group_by(company_name) %>%
  summarise(
    avg_rating = mean(rating_overall, na.rm = TRUE),
```

```r
    n_reviews = n(),
    .groups = "drop"
  ) %>%
  filter(n_reviews >= 5) %>%
  arrange(desc(avg_rating)) %>%
  slice_head(n = 10)

ggplot(top_companies,
       aes(x = reorder(company_name, avg_rating), y = avg_rating)) +
  geom_col(fill = "steelblue") +
  coord_flip() +
  labs(
    title = "Top Companies by Average Rating",
    subtitle = "Among companies with at least 5 reviews",
    x = "Company",
    y = "Average Rating"
  )
```

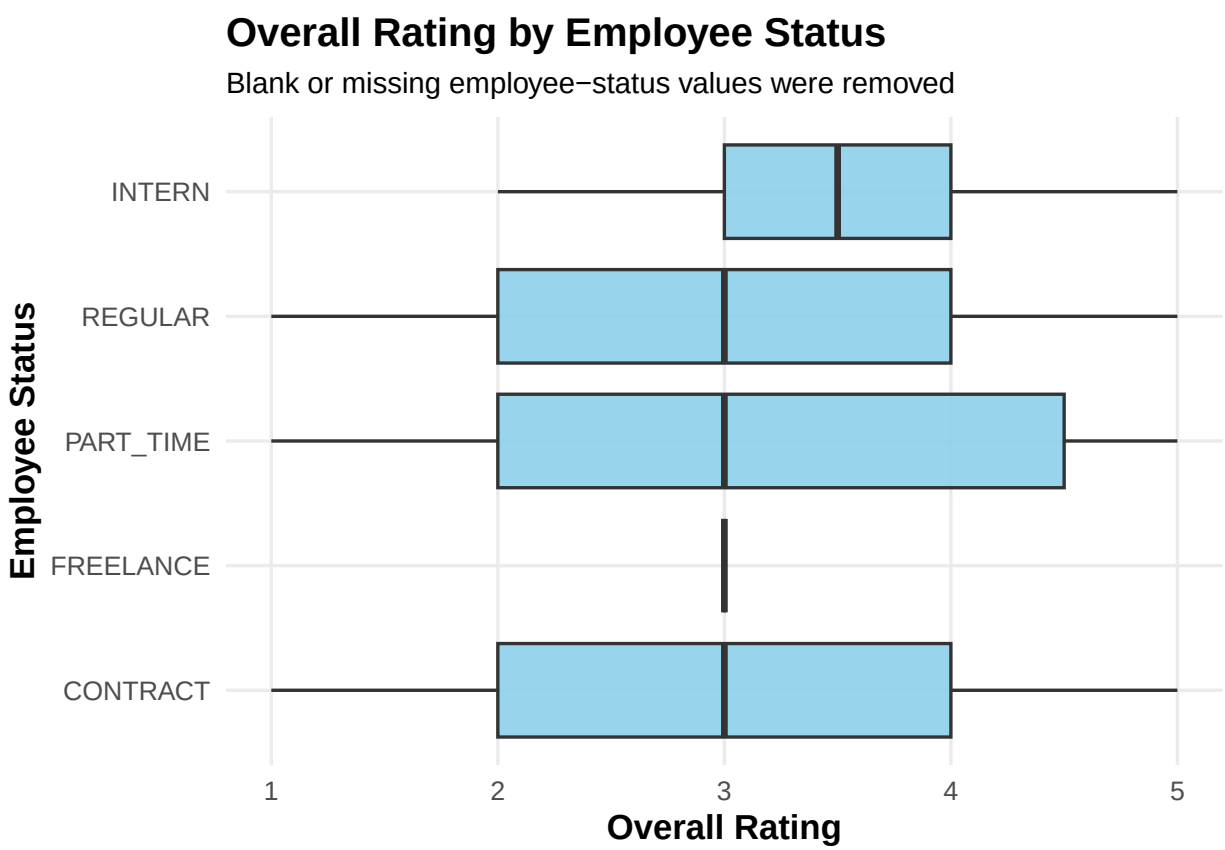


Explanation

The highest-rated companies in the filtered chart represent organizations with both strong average ratings and a minimum number of reviews. Requiring multiple reviews makes the comparison more reliable, since companies with only one or two reviews can appear artificially strong. This plot highlights companies that stand out more credibly in employee satisfaction.

13. Boxplot On Ratings By Employee Status

```r
status_plot_df <- df %>%
  mutate(
    employee_status = trimws(as.character(employee_status)),
    employee_status = ifelse(employee_status == "" | is.na(employee_status),
                             NA, employee_status)
  ) %>%
  filter(!is.na(employee_status))

ggplot(status_plot_df,
       aes(x = fct_reorder(employee_status, rating_overall, .fun = median),
           y = rating_overall)) +
  geom_boxplot(fill = "skyblue", alpha = 0.85, outlier.alpha = 0.35) +
  coord_flip() +
  labs(
    title = "Overall Rating by Employee Status",
    subtitle = "Blank or missing employee-status values were removed",
    x = "Employee Status",
    y = "Overall Rating"
  )
```

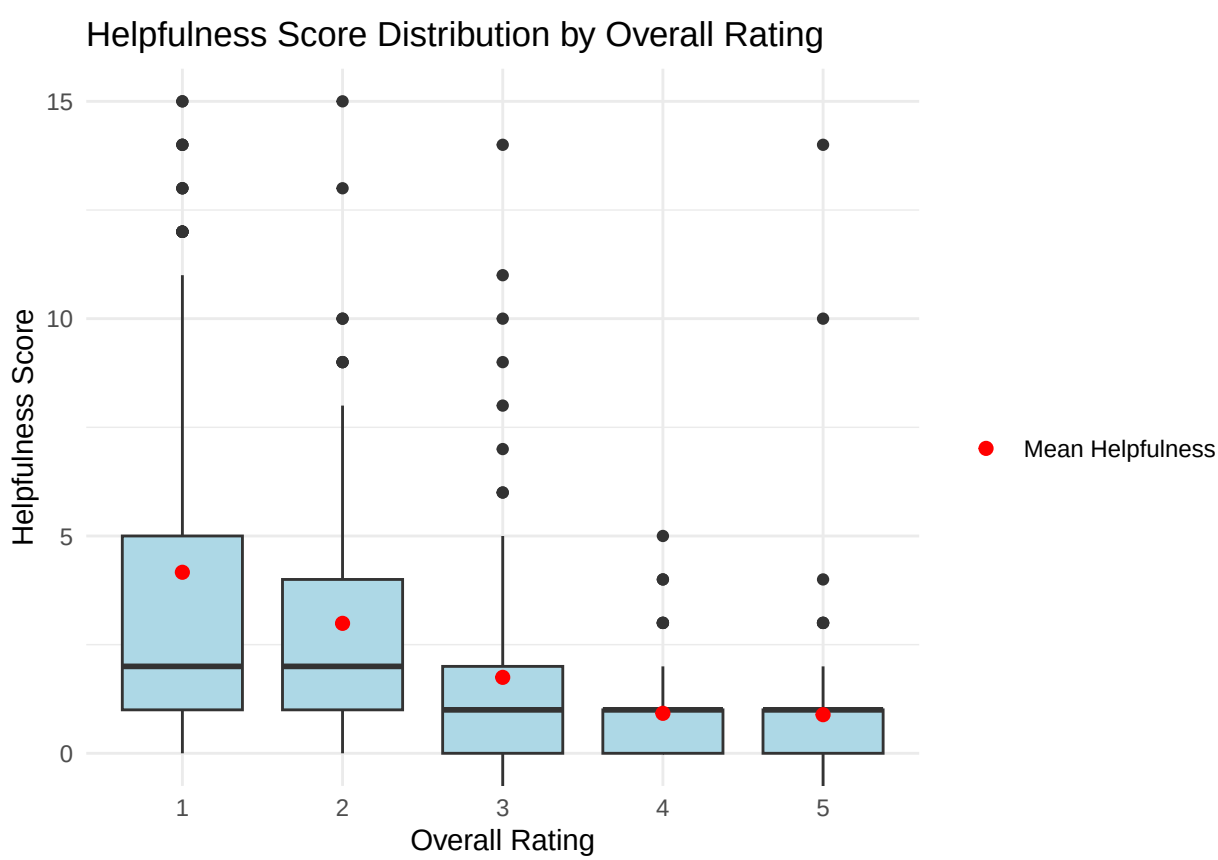


Explanation

Interns report slightly higher satisfaction, while contract employees report somewhat lower ratings. However, the large overlap across groups suggests that employee status is not a major factor in determining overall ratings. The freelance category has very few observations and is not reliable.

14. Boxplot On Helpfulness Score by Overall Rating

```
ggplot(df, aes(x = factor(rating_overall, levels = c(1,2,3,4,5)),
               y = helpfulness_score)) +
  geom_boxplot(fill = "lightblue") +
  stat_summary(aes(color = "Mean Helpfulness"),
               fun = mean, geom = "point", size = 2) +
  scale_color_manual(
    name = "",
    values = c("Mean Helpfulness" = "red")
  ) +
  coord_cartesian(ylim = c(0, 15)) +
  theme_minimal() +
  labs(
    title = "Helpfulness Score Distribution by Overall Rating",
    x = "Overall Rating",
    y = "Helpfulness Score"
  )
```



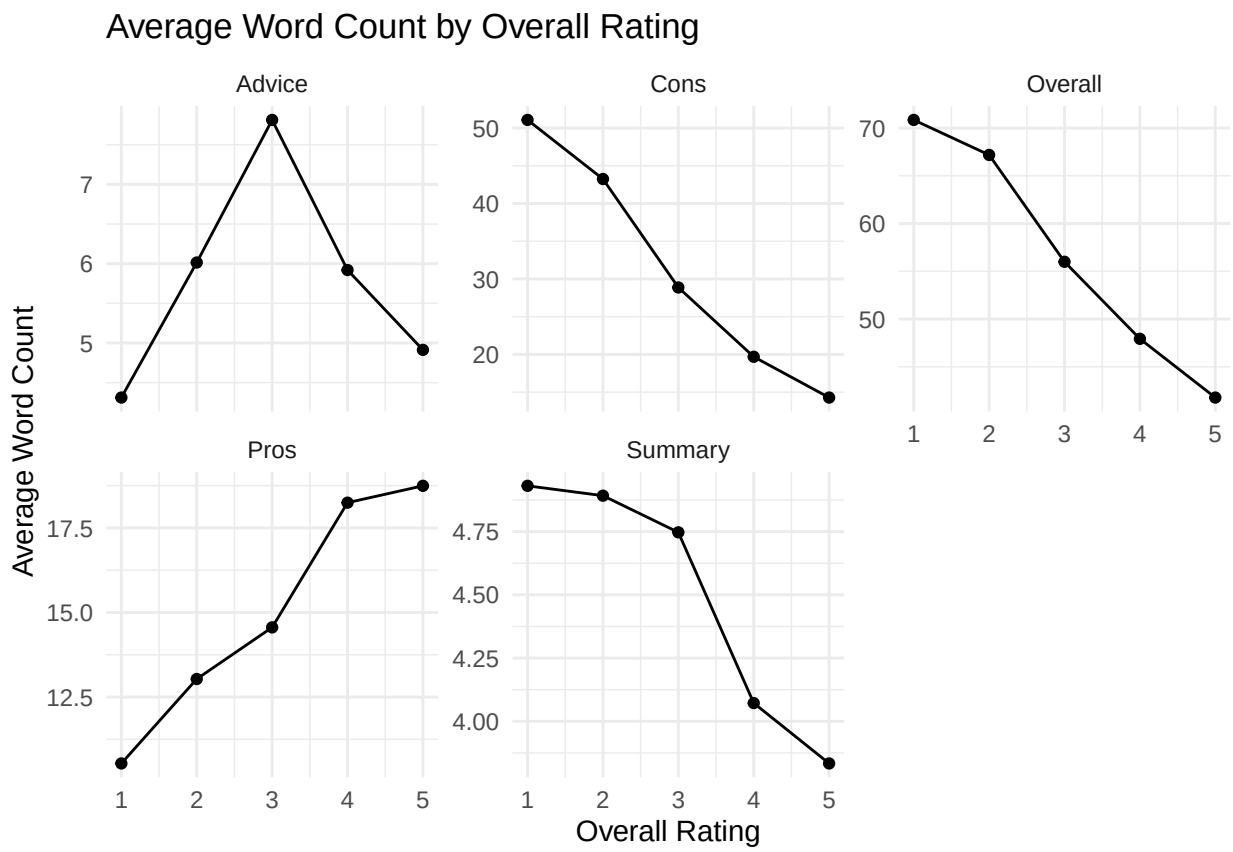
Explanation

Helpfulness scores are highest and most variable for low ratings (1–2), while high ratings (4–5) have consistently low helpfulness. This suggests that negative reviews are more likely to be perceived as informative, whereas positive reviews tend to receive less engagement.

15. Mean Word Count in Review Text Fields by Overall Rating

```
count_words <- function(x) {
  x <- str_squish(x)
  ifelse(is.na(x) | x == "", 0, str_count(x, "\\S+"))
}

wc_rating_df <- df %>%
  mutate(
    summary_wc = count_words(summary),
    pros_wc = count_words(review_pros),
    cons_wc = count_words(review_cons),
    advice_wc = count_words(advice_to_management),
    overall_wc = summary_wc + pros_wc + cons_wc + advice_wc
  )

wc_rating_long <- wc_rating_df %>%
  select(rating_overall, summary_wc, pros_wc, cons_wc, advice_wc, overall_wc) %>%
  pivot_longer(
    cols = c(summary_wc, pros_wc, cons_wc, advice_wc, overall_wc),
    names_to = "column",
    values_to = "word_count"
  ) %>%
  mutate(
    column = recode(column,
      summary_wc = "Summary",
      pros_wc = "Pros",
      cons_wc = "Cons",
      advice_wc = "Advice",
      overall_wc = "Overall"
    )
  )

wc_rating_summary <- wc_rating_long %>%
  group_by(column, rating_overall) %>%
  summarise(mean_wc = mean(word_count, na.rm = TRUE), .groups = "drop")

ggplot(wc_rating_summary, aes(x = rating_overall, y = mean_wc)) +
  geom_line() +
  geom_point() +
  facet_wrap(~ column, scales = "free_y") +
  theme_minimal() +
  labs(
    title = "Average Word Count by Overall Rating",
    x = "Overall Rating",
```

```
    y = "Average Word Count"
  )
```



Explanation

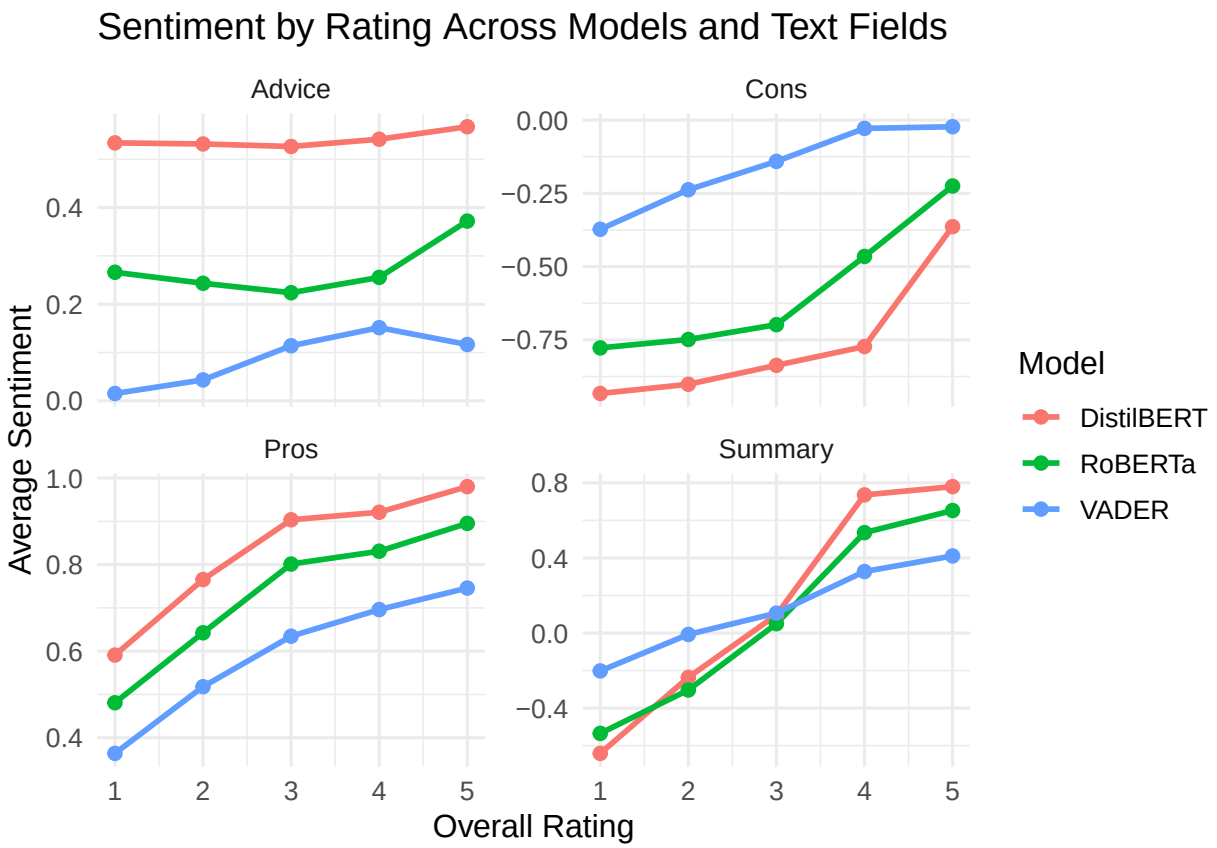
Overall word count decreases steadily as ratings increase, indicating that lower-rated reviews tend to be longer and more detailed. This pattern is primarily driven by the "Cons" section, which declines sharply with higher ratings, while "Pros" increase with rating. This suggests that negative reviews contain more extensive criticism, contributing to both higher word counts and perceived helpfulness.

It is interesting to note that the average word count per across *all* text fields are relatively low (< 70). This fact will be relevant when looking at sentiment score importance during the modeling stage.

16. Sentiment Scores across Text Fields vs Overall Rating

```
sent_long <- df %>%
  select(
    rating_overall,

    # DistilBERT
    distilbert_summary_signed,
    distilbert_pros_signed,
```

```r
    distilbert_cons_signed,
    distilbert_advice_signed,

    # RoBERTa
    roberta_summary_signed,
    roberta_pros_signed,
    roberta_cons_signed,
    roberta_advice_signed,

    # VADER
    vader_summary_compound,
    vader_pros_compound,
    vader_cons_compound,
    vader_advice_compound
  ) %>%
  pivot_longer(
    cols = -rating_overall,
    names_to = "variable",
    values_to = "sentiment"
  ) %>%
  mutate(
    model = case_when(
      grepl("distilbert", variable) ~ "DistilBERT",
      grepl("roberta", variable) ~ "RoBERTa",
      grepl("vader", variable) ~ "VADER"
    ),
    field = case_when(
      grepl("summary", variable) ~ "Summary",
      grepl("pros", variable) ~ "Pros",
      grepl("cons", variable) ~ "Cons",
      grepl("advice", variable) ~ "Advice"
    )
  )

sent_summary <- sent_long %>%
  group_by(model, field, rating_overall) %>%
  summarise(mean_sent = mean(sentiment, na.rm = TRUE), .groups = "drop")

ggplot(sent_summary, aes(x = rating_overall, y = mean_sent, color = model)) +
  geom_line(linewidth = 1) +
  geom_point(size = 2) +
  facet_wrap(~ field, scales = "free_y") +
  theme_minimal(base_size = 12) +
  labs(
    title = "Sentiment by Rating Across Models and Text Fields",
    x = "Overall Rating",
    y = "Average Sentiment",
    color = "Model"
  )
```



Explanation

Sentiment across all text fields exhibits a consistent positive relationship with overall rating, with the most pronounced gradients observed in the "Cons" and "Summary" sections.

The "Cons" field, in particular, shows a strong transition from highly negative to near-neutral sentiment as ratings increase, indicating that negative feedback (or sentiment polarity in general) is a key driver of overall sentiment. While transformer-based models produce sharper separation, VADER captures similar trends, suggesting that sentiment polarity dominates over nuanced contextual differences.